

WS21 - Understanding clValid's Internal validation output

Study the output below and answer the following questions:

```
> library(clValid)
> df <- scale(iris[, -5])
> clmethods <- c("hierarchical", "kmeans", "pam")
>
> intern <- clValid(df, nClust = 2:6,
+                 clMethods = clmethods, validation = "internal")
Warning message:
In clValid(df, nClust = 2:6, clMethods = clmethods, validation = "internal") :
  rownames for data not specified, using 1:nrow(data)
> summary(intern)
```

Clustering Methods:
hierarchical kmeans pam

Cluster sizes:
2 3 4 5 6

Validation Measures:

		2	3	4	5	6
hierarchical	Connectivity	0.9762	5.5964	7.5492	18.0508	24.7306
	Dunn	0.2674	0.1874	0.2060	0.0700	0.0762
	Silhouette	0.5818	0.4803	0.4067	0.3746	0.3248
kmeans	Connectivity	0.9762	23.8151	25.9044	40.3060	40.1385
	Dunn	0.2674	0.0265	0.0700	0.0808	0.0808
	Silhouette	0.5818	0.4599	0.4189	0.3455	0.3441
pam	Connectivity	0.9762	23.0726	31.8067	35.7964	44.5413
	Dunn	0.2674	0.0571	0.0566	0.0642	0.0361
	Silhouette	0.5818	0.4566	0.4091	0.3574	0.3400

Optimal Scores:

	Score	Method	Clusters
Connectivity	0.9762	hierarchical	2
Dunn	0.2674	hierarchical	2
Silhouette	0.5818	hierarchical	2

(a) Is a high or low connectivity score preferred in a clustering result? Is a high or low Dunn index preferred in a clustering result? What about silhouette score – high or low is preferred?

(b) Based only on Connectivity scores, which method(s) (hierarchical, kmeans or pam) and what value of k should you choose as the best method?

(c) Under Optimal Scores, clValid reports the best method as Hierarchical using k = 2. Based on your answer in part (b), can you explain why clValid did not report the following equally valid results as Optimal Scores? Note that the textbook did not offer any explanation.

Optimal Scores:

	Score	Method	Clusters
Connectivity	0.9762	pam	2
Dunn	0.2674	pam	2
Silhouette	0.5818	pam	2

Optimal Scores:

	Score	Method	Clusters
Connectivity	0.9762	kmeans	2
Dunn	0.2674	kmeans	2
Silhouette	0.5818	kmeans	2

- (d) If you have to single out a “best” solution, can you provide sound reasoning which one you will choose? The textbook did not teach you additional useful graphical plots but the `clValid` documentation provided in class (page 13) teaches you how to produce the following plots that might help you choose the “best” solution. You are expected to produce such plots for your term project.

```
> op<-par(no.readonly=TRUE)
> par(mfrow=c(2,2),mar=c(4,4,3,1))
> plot(intern, legend=FALSE)
> plot(nClusters(intern), measures(intern, "Dunn")[,1], type="n", axes=F, xlab="", ylab="")
> legend("center", clusterMethods(intern), col=1:9, lty=1:9, pch=paste(1:9))
> par(op)
```



